# Assignment 9: Travel Budget Analysis

Course: CS101 Computational Thinking | B.Tech 1st Year

## Problem Statement

Convert international travel expenses into a single base currency (INR). Incorporate varying exchange conversion fees, aggregate the total planned spending against a fixed budget, and dynamically identify the largest expense category.

## Algorithm

1. Define a list of tuples containing conversion rates: (Currency Code, Rate to INR).
2. Define expenses: (Category, Amount, Currency Code).
3. Iterate over the expenses. For each, use a nested loop to find the matching currency rate.
4. Apply the conversion arithmetic along with a 2% bank transaction fee.
5. Track the maximum converted expense and its category name.
6. Print the summed totals and identify budget overflows.

## Python Implementation

```
# (Currency, Value in INR)
exchange_rates = [("USD", 83.5), ("EUR", 89.2), ("GBP", 104.1), ("INR", 1.0)]

# (Category, Amount, Currency)
expenses = [
    ("Accommodation", 600, "USD"),
    ("Food", 400, "EUR"),
    ("Internal Flights", 15000, "INR"),
    ("Sightseeing", 250, "GBP")
]

budget_inr = 120000
total_spent_inr = 0.0

highest_expense_cat = ""
highest_expense_val = 0.0

for exp in expenses:
    category, amount, curr = exp
    rate = 0.0

    # Nested iteration to simulate basic lookup without dictionaries
    for r in exchange_rates:
        if r[0] == curr:
            rate = r[1]

    # Calculate converted value + 2% forex markup fee
    converted_val = amount * rate
    if curr != "INR":
        converted_val *= 1.02

    total_spent_inr += converted_val

    if converted_val > highest_expense_val:
        highest_expense_val = converted_val
        highest_expense_cat = category
```